ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 4

ROBOTICS AND INDUSTRIAL AUTOMATION

PAPER 2 · PRACTICE A

MARKS: 100 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 4. Paper 2 (design related). Section A: objective questions (30) · Section B: industrial robots, grippers and end-of-arm tooling, gripper physics, networks and industrial communication, smart industry and cyber security, and a robot work-cell design task (70).

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 4. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Show ALL formulae, substitutions and calculations, and write units with all answers. Round off to TWO decimal places.
5. Use g = 9,81 m/s² where needed.
6. A non-programmable calculator may be used.
7. Write neatly and legibly.

SECTION A

QUESTION 1

1.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1.1–1.1.10).

1.1.1 Robots are often chosen for jobs that are dull, dirty and …

A delicate. B digital.
C distant. D dangerous.

1.1.2 The international organisation that publishes the standard definition of an industrial robot is …

A SABS. B ISO. C IEEE only. D DHET.

1.1.3 The space that a robot's end effector can reach is called the …

A payload. B footprint.
C work envelope. D cycle time.

1.1.4 EOAT stands for …

A end-of-arm tooling. B electric output and torque.
C emergency override and trip. D external operating arm table.

1.1.5 A gripper that lifts flat sheets of glass using suction is a …

A needle gripper. B vacuum cup gripper.
C 3-jaw gripper. D magnetic gripper.

1.1.6 A network that covers a whole city is a …

A PAN. B LAN. C WAN only. D MAN.

1.1.7 In a … topology, every device is connected to one central switch.

A star B bus C ring D mesh

1.1.8 A communication system that links many field devices (sensors, valves, drives) to a controller on one cable is a …

A power bus. B USB hub.
C fieldbus. D telephone line.

1.1.9 IIoT stands for …

A Integrated Industrial Output Terminal. B Internal Input over TCP.
C Industrial Internet of Things. D Intelligent Interface of Tools.

1.1.10 Processing data on or near the device, instead of in the cloud, is called …

A edge processing. B cloud storage.
C batch processing. D offline programming.

(10)

1.2 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'true' or 'false' next to the question number (1.2.1–1.2.10).

1.2.1 A SCARA robot is very rigid in the vertical direction and flexible in the horizontal plane.

1.2.2 Repeatability is the robot's ability to return to the same point time after time.

1.2.3 Hydraulic robots are best for very light, clean pick-and-place tasks.

1.2.4 The available payload is the rated payload minus the mass of the gripper and tooling.

1.2.5 Friction force increases when the gripping (normal) force increases.

1.2.6 A router connects different networks and forwards data between them.

1.2.7 MODBUS and PROFIBUS are examples of industrial communication protocols.

1.2.8 Industrial Ethernet cannot be used on a factory floor.

1.2.9 A work cell is a group of machines, robots and equipment that work together on one task.

1.2.10 Cyber security is not important for IoT devices because they are small.

(10)

1.3 Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–J) next to the question number (1.3.1–1.3.10).

	COLUMN A		**COLUMN B**
1.3.1	Robot with parallel arms for very fast picking	A	MIG welder
1.3.2	Area of floor that the robot base takes up	B	safety factor
1.3.3	Welding tool that feeds wire through a gun with shielding gas	C	4D printing
1.3.4	Turning effect of a force	D	peer-to-peer
1.3.5	Extra margin used when calculating gripping force	E	torque
1.3.6	Device that connects devices inside one LAN	F	switch
1.3.7	Network where all computers are equal and share directly	G	delta robot

1.3.8	Set of rules for sending data	H	lead-through programming
1.3.9	Printing objects that change shape after printing	I	protocol
1.3.10	Moving the robot by hand to record the path	J	footprint

(10)

TOTAL SECTION A: [30]

SECTION B

QUESTION 2: INDUSTRIAL ROBOTS AND GRIPPERS

2.1 Robots are often used for jobs that are dull, dirty or dangerous. Explain EACH of these three words and give an industrial example of each, then give ONE other reason (besides the three D's) why a company may choose a robot. (4)

2.2 Compare a SCARA robot and a six-axis articulated robot under the headings articulation, work envelope, payload and typical application. (8)

2.3 Name the most suitable gripper for EACH object and give a reason: (a) steel plates; (b) soft fruit; (c) round shafts of different sizes. (6)

2.4 Differentiate between point-to-point and continuous-path control, and name ONE application of each. (4)

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QUESTION 3: GRIPPER PHYSICS

3.1 A robot has a rated payload of 10 kg. Its gripper and tool changer have a mass of 2,4 kg.

3.1.1 Calculate the available payload. (2)

3.1.2 Can the robot lift a part with a mass of 8 kg? Motivate your answer. (1)

3.2 The robot lifts a 6 kg box that is held 0,4 m from the wrist axis.

3.2.1 Calculate the weight of the box. (2)

3.2.2 Calculate the torque the box produces about the wrist. (2)

3.3 A two-jaw gripper holds a 2 kg part by friction (vertical lift). The coefficient of friction between the jaws and the part is 0,4. A safety factor of 2 must be used.

3.3.1 Calculate the weight of the part. (2)

3.3.2 Calculate the total normal (gripping) force needed so that friction holds the part, including the safety factor. (4)

[13]

QUESTION 4: NETWORKS AND INDUSTRIAL COMMUNICATION

4.1 Differentiate between client-server and peer-to-peer network architecture. (4)

4.2 Give ONE advantage and ONE disadvantage of a star topology and of a bus topology. (4)

4.3 State the function of EACH of the following networking devices: gateway, router, wireless access point. (3)

4.4 Give FOUR reasons why factories use industrial Ethernet. (4)

4.5 Give TWO advantages and ONE disadvantage of fieldbus systems compared with wiring every device directly to the controller. (3)

[18]

QUESTION 5: SMART INDUSTRY AND DESIGN

5.1 Explain the relationship between the cloud and the IoT, and give ONE example of how the IIoT is used where robots work. (3)

5.2 Name FOUR ways to protect a robot work cell's network from cyber attacks. (4)

5.3 A factory wants a robot cell that takes bottles from a conveyor and packs them into boxes of six. Design the cell by answering the following:

5.3.1 Recommend a robot type and a gripper type, with reasons. (4)

5.3.2 Name FOUR sensors or safety devices the cell needs and the purpose of each. (4)

5.3.3 Name TWO aspects that must be considered when integrating the robot with the existing conveyor and factory network. (2)

[17]

TOTAL SECTION B: [70]

TOTAL: 100

END OF QUESTION PAPER